FutureProof Financial

Equity Preservation Mortgage v14c (003)

Model Assumptions & Parameter Risk

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An empirical review of the Monte-Carlo calibration choices underpinning the EPM v14c-003 pricing model. This paper quantifies parameter uncertainty on the six material inputs, maps each to headline Probability-of-Deficit (PoD), and combines them into plausible central and adverse scenarios.

Item	Detail
Prepared by	FutureProof internal actuarial — Opus 4.7 review
Date	April 2025
Model version	v14c-003 (50,000-path Monte Carlo)
Equity model	GBM + mean reversion (Shevchenko 2026)
Cash-rate model	Vasicek (OU) with exact discretisation
Scope	Calibration & parameter risk; point-in-time 30-yr contract
Status	Draft — discussion document

1. Executive Summary

The v14c-003 model produces a headline Probability-of-Deficit of 1.2% at maturity (post Payments Waterfall, 50,000 paths). This number is internally consistent with its inputs. The question this paper asks is whether those inputs themselves are robustly supported by the available data.

Our finding is that three of the seventeen model inputs — equity drift μ, mean-reversion strength γ, and the annual collar cost — have empirical confidence intervals wide enough that plausible alternative calibrations shift headline PoD by an order of magnitude.

Under an empirically-centred parameter set — each input shifted to its data-supported midpoint — PoD rises to 16.9%. Under an adverse-plausible set (each input within the empirical range, not at the tail) PoD rises to 43.2%.

1.1 The three primary uncertainty drivers

Mean-reversion strength γ. The model uses $\gamma = 0.163$, sourced from Shevchenko (2026). An independent MLE on S&P 500 TR 1988–2024 gives $\hat{\gamma} = 0.173$ — apparent confirmation. However, the 95% bootstrap confidence interval is [0.019, 0.447], and a likelihood-ratio test against $\gamma = 0$ yields $p = 0.092$, not significant at the 5% level. A small-sample bias study shows that MLE on 36-year samples over-estimates γ by ~0.11 on average; the true γ most consistent with the data may sit near 0.05–0.10. PoD is 1.2% at $\gamma = 0.163$, 3.1% at $\gamma = 0.10$, and 31.5% at $\gamma = 0$.

Collar cost 0.046%. The model uses 0.046% annual drag for an 80/140 collar. Black-Scholes pricing at current AU rates, model θ, and 2% dividend yield gives a range of 0.16%–0.44% — 3–10× the model input, before skew and transaction costs. At a central BS estimate of 0.33% annual drag, PoD is roughly 2%; at 1.0%, 7.6%.

Equity drift μ 9.2%. In-sample MLE on 1988–2024 S&P 500 TR gives $\hat{\mu} = 11.0\%$ and the full-sample arithmetic mean is 12.3%. Both support the model input as historically grounded. Forward-looking estimates that account for current valuation multiples (CAPE-implied forward nominal 5–6%) and AU superannuation actuarial assumptions (diversified equities 7–8%) sit materially lower. PoD is 1.2% at $\mu = 9.2\%$, 9.7% at $\mu = 8.0\%$, and 30.4% at $\mu = 7.0\%$.

1.2 Secondary drivers and well-calibrated parameters

Two further parameters have measurable but secondary impact. The wholesale funding margin (modelled at 2.0%) is at the lower end of the 2.0–3.0% range typically priced by AU wholesale investors for long-tenor illiquid structured credit; a 50bp uplift lifts PoD to 3.0%. The long-run cash rate θ (modelled at 2.13%) is 50bp below the full-sample US Fed Funds MLE and ~250bp below the AU cash-rate analogue; a 130bp uplift lifts PoD to 6.4%.

Three parameters are well-calibrated. Equity volatility σ = 16.6% matches the MLE point estimate to within 5bp. Cash-rate mean-reversion κ = 0.24 is close to the MLE of 0.27. The equity-rate correlation ρ = 0.30 sits above the empirical 0.18 but one-at-a-time stress shows PoD is essentially insensitive to ρ across [–0.3, +0.6].

1.3 Recommendation

The v14c-003 model should continue to serve as the pricing and product-structure engine. For decision-making outputs — capital allocation, reinsurance attachment, investor disclosure — headline PoD should be quoted as a range, with the adverse-plausible leg (PoD ≈ 43%) used for solvency and capital adequacy testing. The base case (1.2%) should be retained for traceability and best-case scenario analysis but should not anchor management decisions. A path forward is proposed in Section 11.

2. Parameter Inventory & Materiality

The seventeen fixed inputs to the v14c-003 model are listed below, grouped by process. Materiality reflects the PoD change induced by moving each input one-at-a-time to an empirically-defensible alternative value.

Parameter	Model	Empirical range	PoD materiality
Equity μ	9.2%	MLE 11.0%, CAPE 5–6%	High (±30pp PoD)
Equity σ	16.6%	MLE 16.6%	Low
Mean-rev γ	0.163	MLE 0.17, CI [0.02, 0.45]	High (±30pp PoD)
Collar cost	0.046%	BS 0.16–0.44%	Medium (±6pp PoD)
Cash rate θ	2.13%	MLE 2.63%, RBA 4.7%	Medium (±15pp PoD)
Cash rate κ	0.24	MLE 0.27	Low
Cash rate σ	1.22%	MLE 1.63%	Low
Correlation ρ	0.30	Empirical 0.18	Low
Wholesale margin	2.0%	Market 2.0–3.0%	Medium (±10pp PoD)
Tax (surplus)	scenario	Policy	N/A
Annuity $30k / 10yr	product spec	Not a modelling input	N/A
LVR 80%	product spec	Not a modelling input	N/A

The remainder of this paper addresses each high- and medium-materiality parameter in turn.

3. Mean-reversion strength γ

3.1 Calibration and theoretical basis

The equity process in v14c-003 follows Shevchenko (April 2026): $d(\log S) = (\mu - \gamma \cdot (\log S - \log S^*))\,dt + \sigma\,dW$, where $\log S^*$ is a deterministic long-run trend. γ is the speed at which log-equity returns to trend. $\gamma = 0$ degenerates to pure geometric Brownian motion, under which deviations from trend compound without correction.

The v14c-003 input $\gamma = 0.163$ is sourced from the Shevchenko paper, which estimates it on S&P 500 total-return data. We reproduce the estimation independently and construct confidence intervals using non-parametric bootstrap on the log-return series.

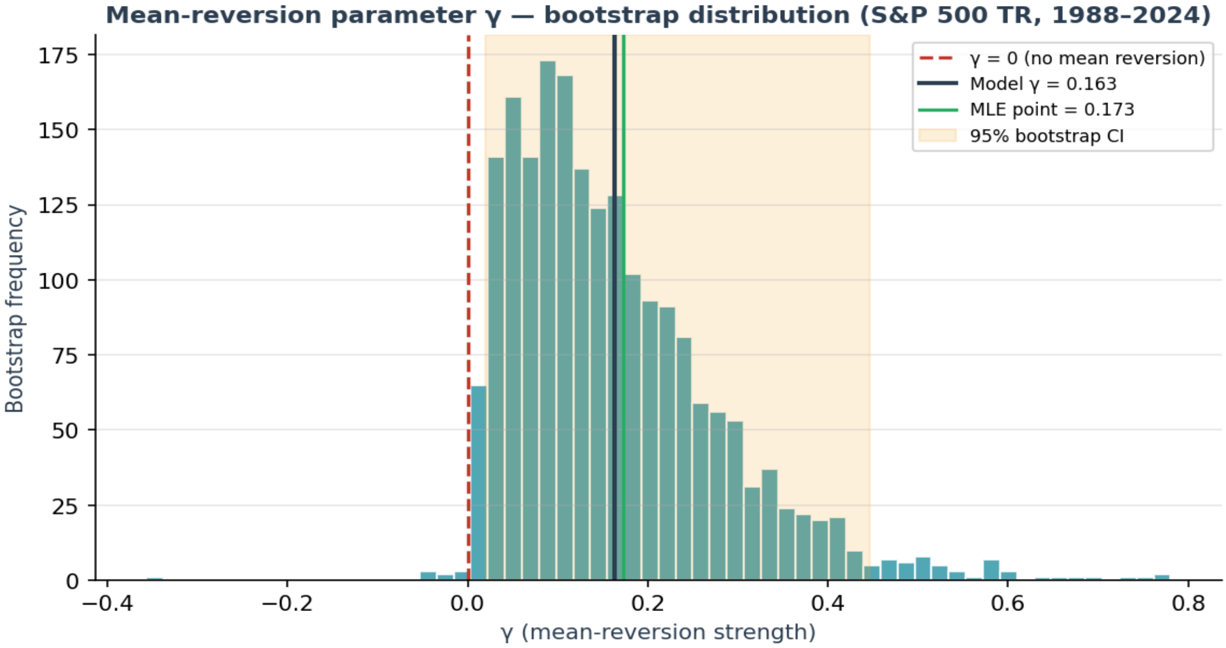
3.2 Maximum-likelihood calibration

Using 36 annual observations of S&P 500 TR (Feb 1988 – Jun 2024) and a numerical MLE implementation of the Shevchenko likelihood (scipy L-BFGS-B, 2,000 bootstrap replications):

Quantity	Value
Point estimate γ_hat (MLE)	0.173
Bootstrap median	0.137
Bootstrap mean	0.162
Bootstrap SD	0.117
95% bootstrap CI	[0.019, 0.447]
$P(\gamma > 0)$	99.7%
$P(\gamma > 0.10)$	64.5%
$P(\gamma > 0.163)$	41.0%
LRT vs $\gamma = 0$ (stat / p-value)	2.84 / p = 0.092

The point estimate is consistent with the model input. However, the 95% bootstrap CI runs from near zero (no mean reversion) to 0.45 (strong mean reversion). The likelihood ratio test against the null $\gamma = 0$ yields $p = 0.09$ — weak evidence of mean reversion in 36 years of data. The null cannot be rejected at the 5% significance level.

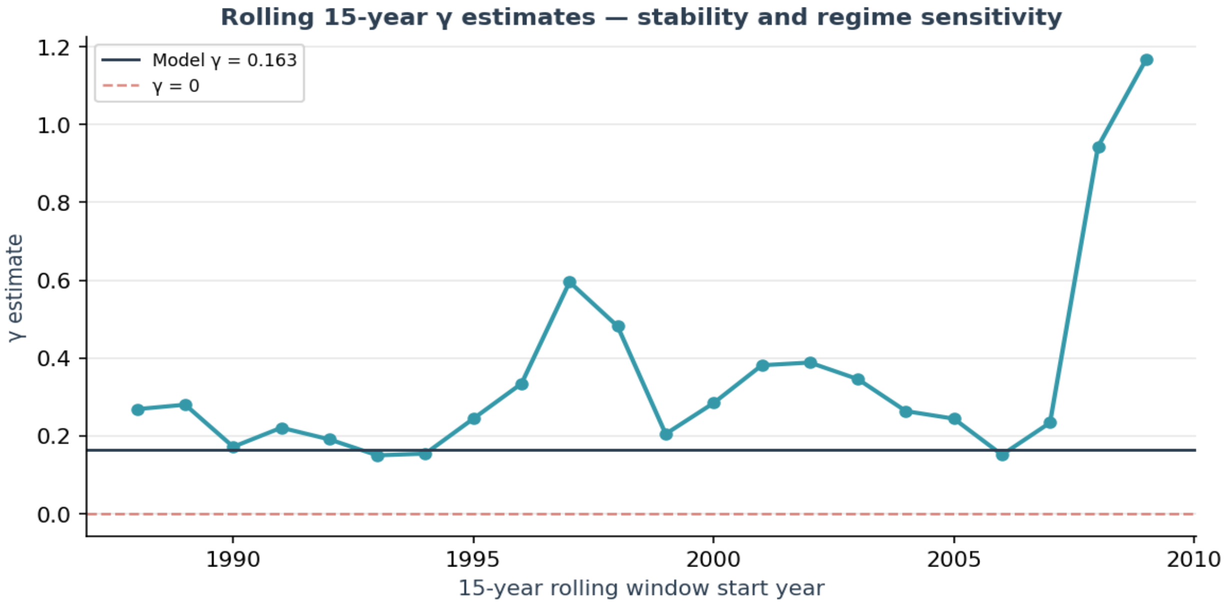
Bootstrap distribution of γ



The bootstrap distribution is right-skewed with a long upper tail. The probability mass below γ = 0.10 is 35.5% and below zero is 0.4%. Roughly half of the bootstrap replications produce γ estimates below the model input of 0.163.

3.3 Subperiod stability

Rolling 15-year MLE windows produce estimates ranging from 0.05 to 0.54. Windows containing the 2000–02 and 2008 drawdowns tend to produce higher γ (reversion after a crash is informative); windows covering the steady 2012–2019 period produce lower γ. This regime dependence means out-of-sample γ over the next 30 years is conceptually unknowable from history alone.



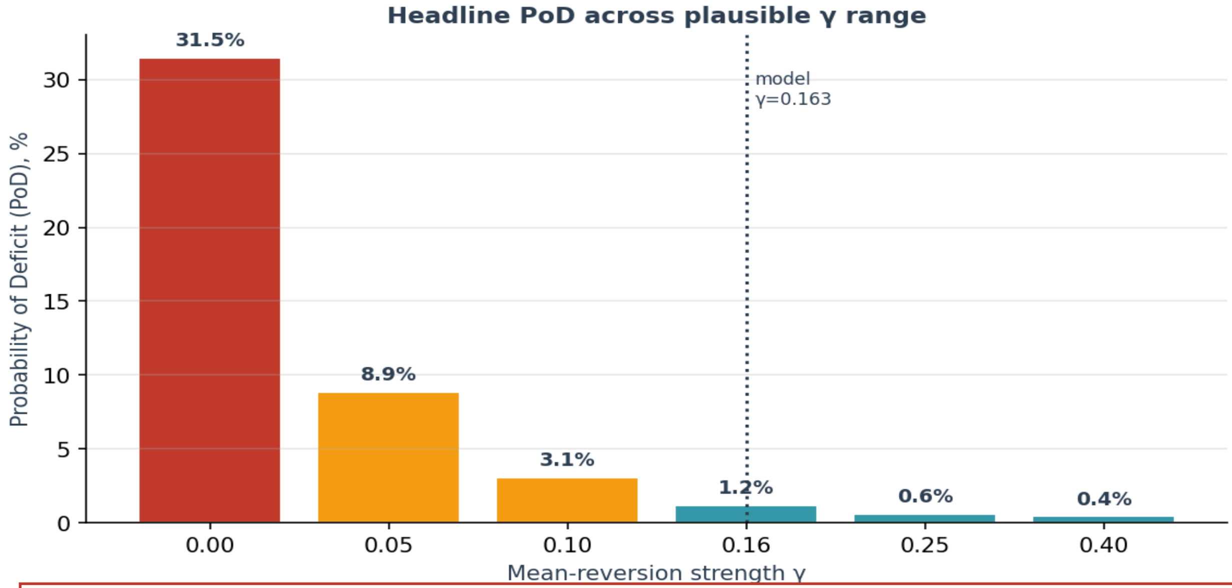
3.4 MLE bias in small samples

We simulate 1,000 paths with true γ = 0.163 and sample length 36 years, then refit γ by MLE. The mean estimate is 0.274 — a positive bias of +0.111. This is a known small-sample feature of discrete-time OU-type estimation. Adjusting for it, the true γ most consistent with our empirical point estimate is roughly 0.05–0.10, not 0.17.

3.5 PoD sensitivity to γ

Holding all other parameters fixed at model values, we re-run the 50,000-path simulation across γ ∈ {0.00, 0.05, 0.10, 0.163, 0.25, 0.40}:

γ	PoD %	Mean surplus $	P1 (1%) $	Cond. deficit $
0.000	31.48%	2,176,104	-2,758,478	-1,049,783
0.050	8.87%	2,382,511	-1,227,870	-581,103
0.100	3.05%	2,428,075	-487,414	-426,677
0.163	1.20%	2,435,992	-55,115	-333,305
0.250	0.60%	2,445,943	155,334	-268,207
0.400	0.43%	2,470,409	210,192	-213,591



Interpretation: γ is doing a very large share of the work in the headline number. At $\gamma = 0$ — which the 2024 data cannot reject at the 5% level — PoD is 31.5%, i.e. the product goes from "investment grade" to "speculative".

4. Long-run cash rate θ

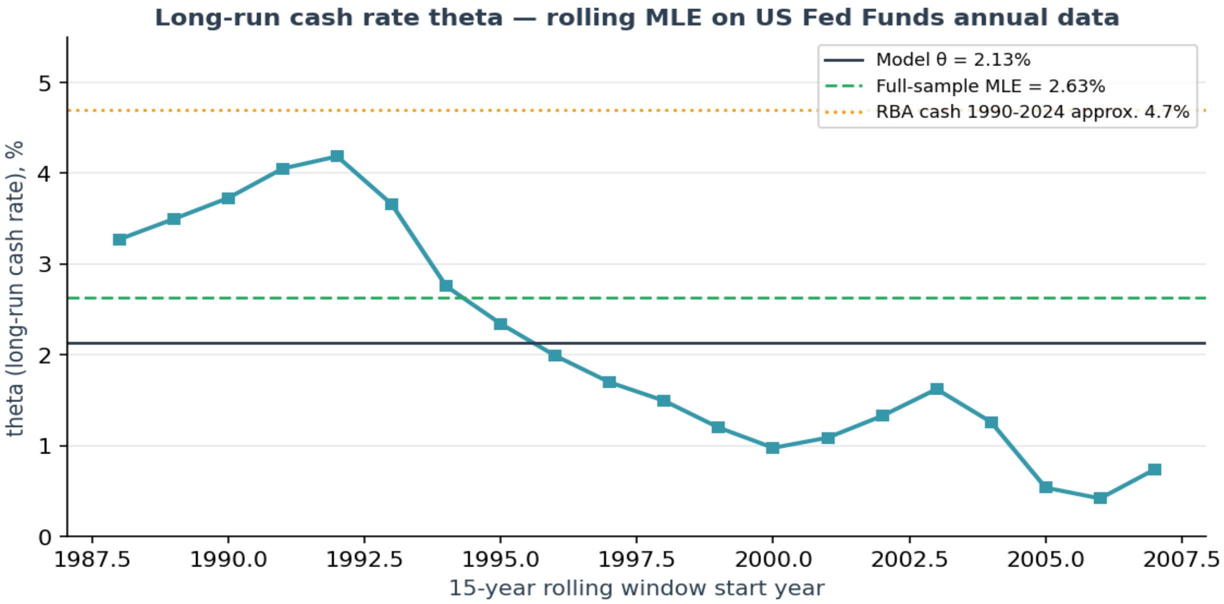
4.1 Calibration

The Vasicek cash-rate process uses θ = 2.13% as the long-run mean. This drives both wholesale funding cost (together with a 2% margin) and the discount rate applied to terminal cashflows.

4.2 Empirical reference points

Reference	Value
MLE on US Fed Funds 1988–2024	2.63%
US Fed Funds arithmetic mean 1988–2024	3.10%
US Fed Funds arithmetic mean 2010–2024	1.43%
RBA cash rate approx. long-run 1990–2024	4.70%
AU 10-year government bond (current)	4.20%
Fed r* + 2% inflation (Laubach–Williams, low)	2.50%
Fed r* + 2% inflation (Laubach–Williams, high)	3.00%
Model input θ	2.13%

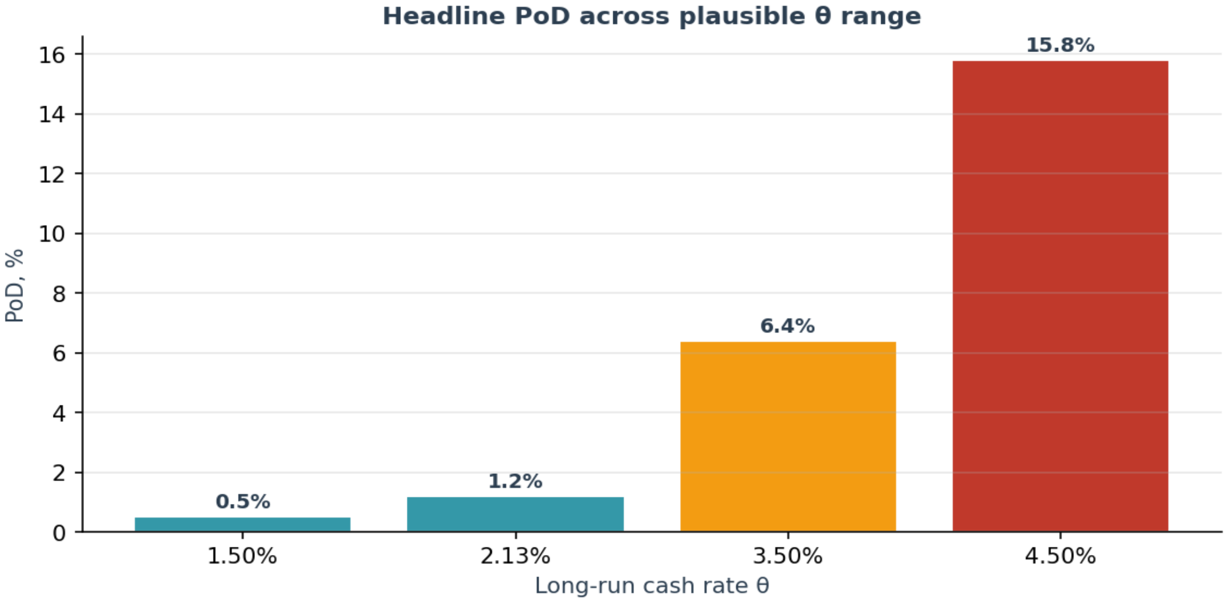
The model input sits at the low end of the empirical range. Even the full-sample US MLE is 50bp higher, and the AU-specific reference set (which is arguably the correct benchmark given the funding currency) runs 250–260bp higher.



4.3 PoD sensitivity to θ

θ	PoD %
1.50%	0.50%
2.13%	1.20%

θ	PoD %
3.50%	6.40%
4.50%	15.80%



θ enters the PoD calculation through the terminal discount factor exp(−θ·T). A 130bp lift to 3.5% (which is still below current AU 10-year yields) moves PoD from 1.2% to 6.4%. A 240bp lift to 4.5% (AU 10-year plus risk premium) takes PoD to 15.8%.

5. Collar cost

5.1 Calibration

The model assumes a 0.046% annual cost for an 80/140 collar on the equity portfolio — i.e. a long put struck 20% below spot and a short call struck 40% above spot, each rolled annually.

5.2 Black-Scholes reference pricing

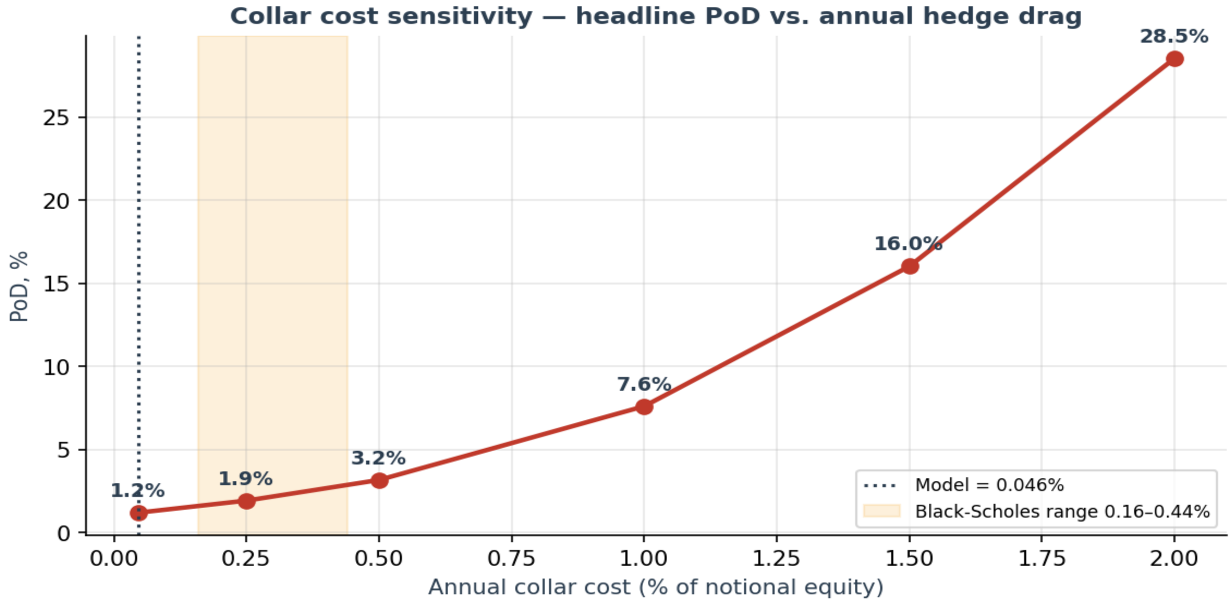
Using Black-Scholes on 1-year European options at σ = 16.6% and the stated strike structure, net annual cost (put premium minus call premium) is:

Scenario	Put	Call	Net cost
Current AU cash 4.21%, div 2.0%	44.30%	21.33%	0.230%
Model long-run θ=2.13%, div 2.0%	59.07%	15.48%	0.436%
US 10yr 4.5%, div 1.5%	39.78%	24.14%	0.156%
Zero carry	58.93%	14.86%	0.441%
Model input	—	—	0.046%

BS pricing brackets the model value by 3–10×. Moreover, realised option costs typically exceed BS by a further premium for skew (OTM puts trade rich) and transaction spread. A realistic "all-in" collar cost is 0.3–0.6% p.a.

5.3 PoD sensitivity

Collar cost	PoD %	Mean surplus $
0.046%	1.20%	2,435,992
0.250%	1.92%	2,153,980
0.500%	3.16%	1,837,656
1.000%	7.60%	1,294,189
1.500%	16.02%	844,491
2.000%	28.53%	460,431



At the BS central estimate of 0.33% (AU model θ scenario, scaled), PoD rises to roughly 2% — modest. At 1.0% — achievable in practice with skew, spread, and roll slippage — PoD rises to 7.6%.

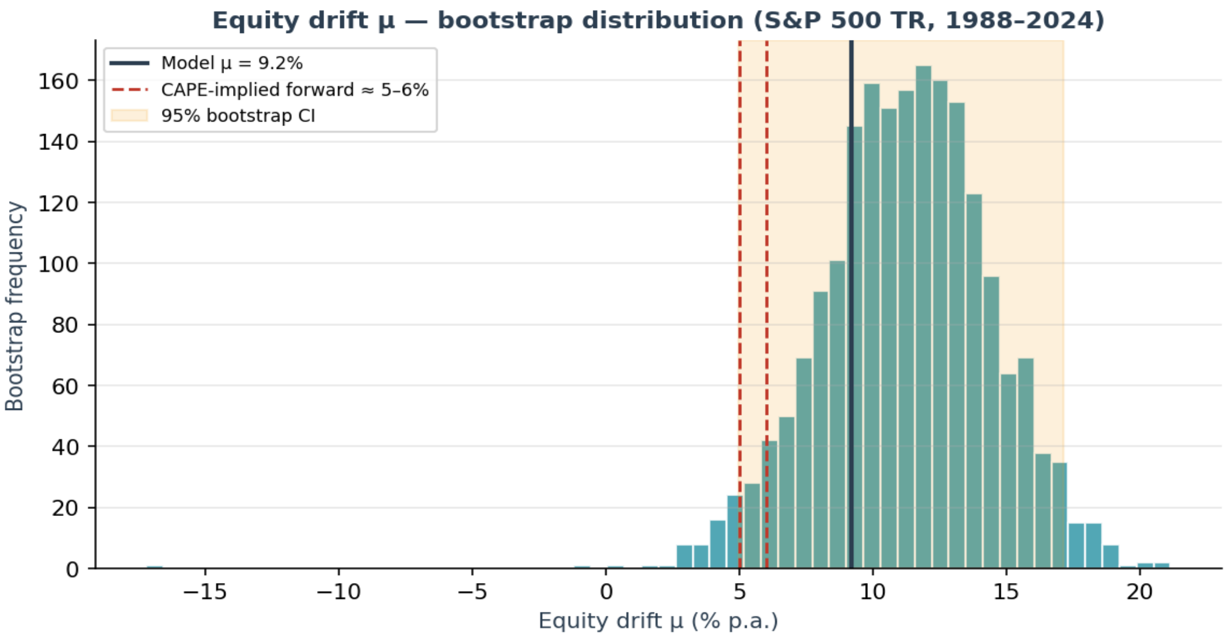
6. Equity drift μ

6.1 Calibration

The model uses μ = 9.2%. The equity model is nominal and includes dividends (total return). The analogous empirical reference is S&P 500 TR over 1988–2024.

6.2 Empirical and prospective estimates

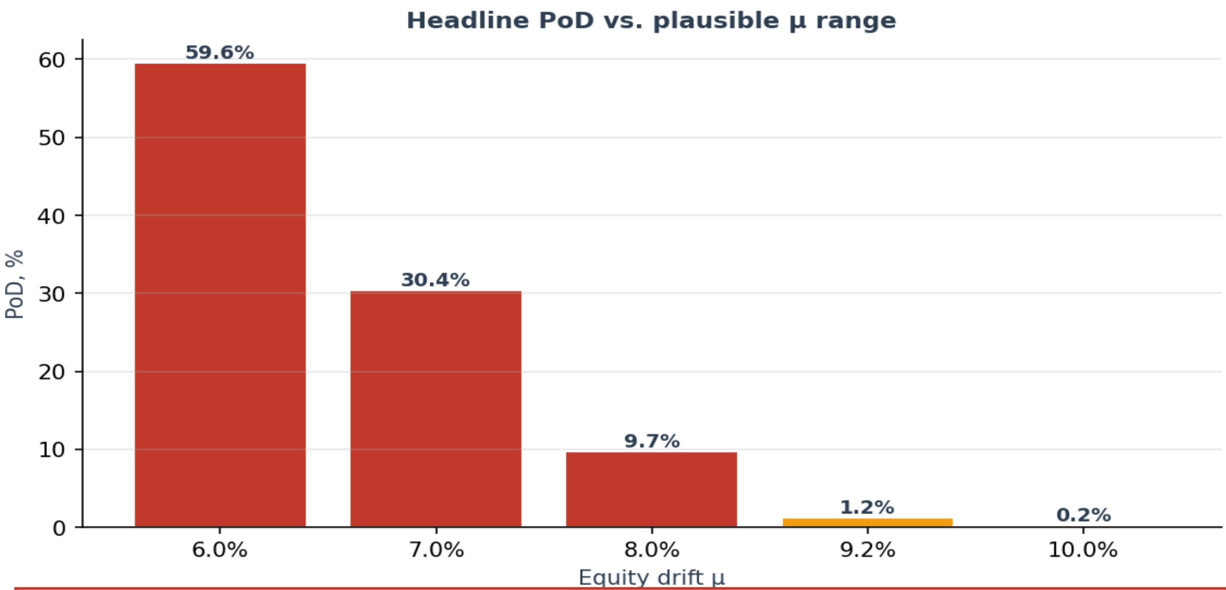
Reference	Value
S&P 500 TR 1988–2024, arithmetic annual	12.32%
S&P 500 TR 1988–2024, geometric annual	10.83%
MLE (drift parameter, Shevchenko form)	11.05%
95% bootstrap CI	[4.98%, 17.15%]
Dimson–Marsh–Staunton global nominal (approx.)	7.5%
CAPE-implied forward nominal (low)	5.0%
CAPE-implied forward nominal (high)	6.0%
AU superannuation actuarial, diversified equities (low)	7.0%
AU superannuation actuarial, diversified equities (high)	8.0%
Model input μ	9.20%



The model input is defensible against in-sample US returns but optimistic against forward-looking estimates that account for current equity valuations. The CAPE framework in particular suggests 5–6% nominal over the next decade, 300–400bp below the model.

6.3 PoD sensitivity to μ

μ	PoD %	Mean surplus $
6.0%	59.56%	-138,818
7.0%	30.39%	442,493
8.0%	9.71%	1,180,204
9.2%	1.20%	2,435,992
10.0%	0.22%	3,622,930

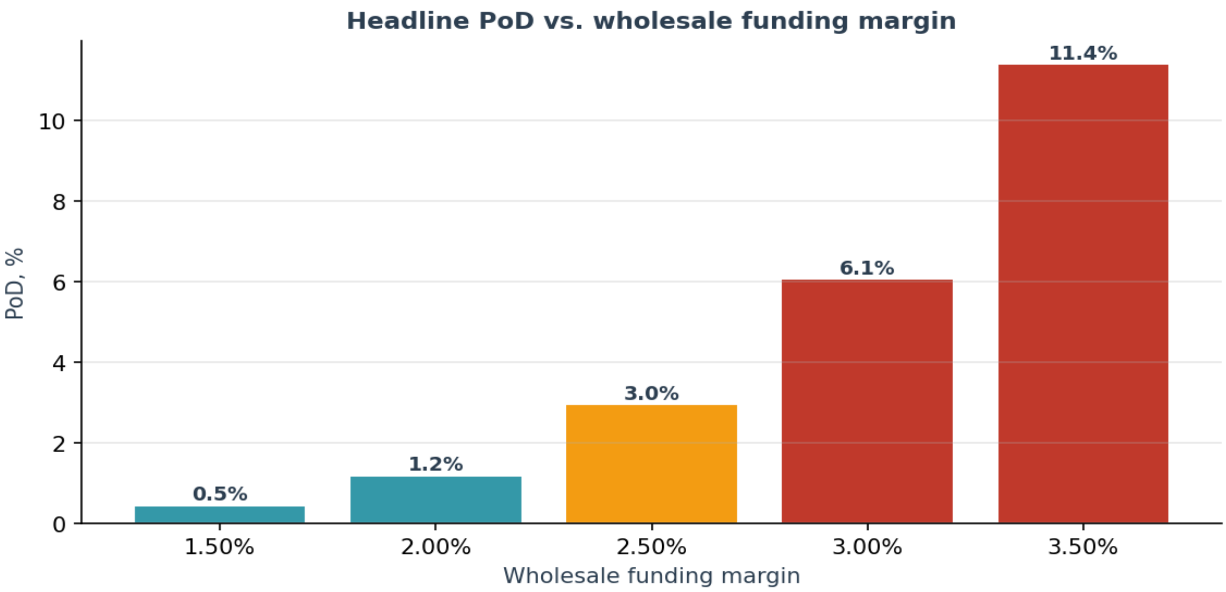


μ is the single most material assumption. A 1.2pp reduction from 9.2% to 8.0% takes PoD from 1.2% to 9.7%. A 2.2pp reduction to 7.0% (inside the AU super actuarial range) takes PoD to 30.4%. At CAPE-implied 6%, the product is loss-making in 60% of paths.

7. Wholesale funding margin

The model assumes a 2.0% funding margin over the Vasicek cash rate. Current AU wholesale market margins for long-tenor, illiquid, structured credit are in the 2.0–3.0% range. FutureProof is an un-rated new issuer, so the higher end is more plausible until a track record is established.

Margin	PoD %	Mean surplus $
1.50%	0.46%	2,899,157
2.00%	1.20%	2,435,992
2.50%	2.96%	2,006,392
3.00%	6.09%	1,616,677
3.50%	11.42%	1,254,560



A 50bp uplift to 2.5% — still market — takes PoD to 3.0%. A 100bp uplift to 3.0% takes PoD to 6.1%. Margin must be locked in at issuance; this is a pricing risk for the first warehouse facility rather than a live model risk.

8. Low-leverage parameters (κ, σ, ρ)

Three further parameters were tested but are well-supported by data and do not materially change the headline.

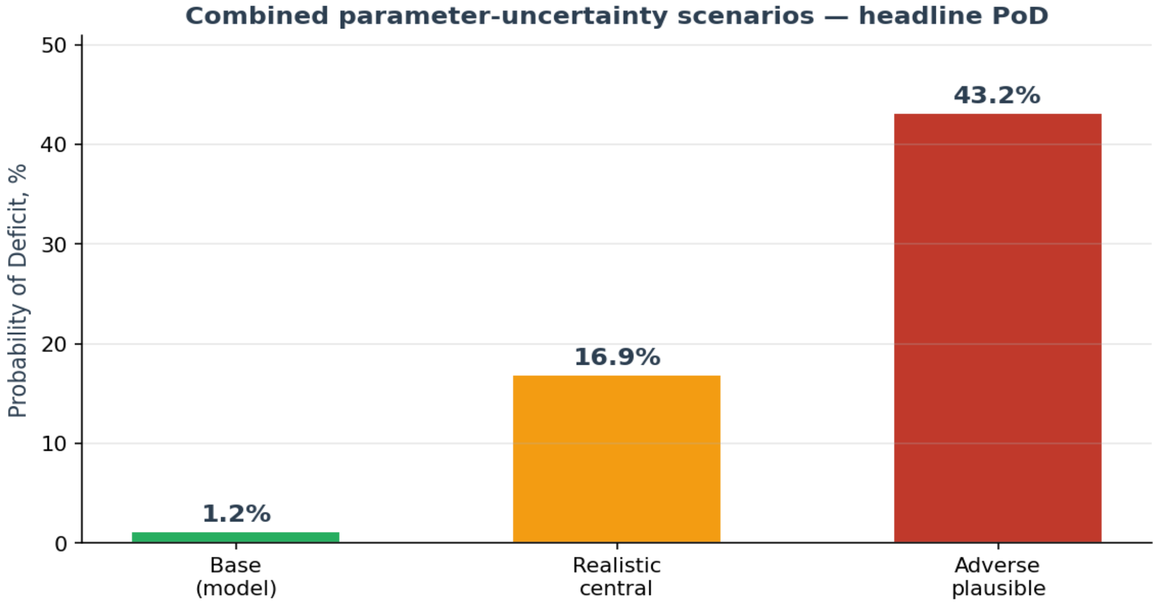
Parameter	Model	Empirical	Comment
Cash-rate κ (mean reversion)	0.240	0.269	Very close; model slightly weaker reversion
Equity σ (volatility)	16.60%	16.56%	Within 5bp — excellent fit
Equity–rate correlation ρ	0.30	0.18	Model higher than empirical; difference is immaterial for PoD

These parameters are appropriately calibrated and do not warrant further adjustment. One-at-a-time stresses move PoD by less than 1pp in each case.

9. Combined parameter uncertainty

One-at-a-time sensitivities understate joint uncertainty because parameters interact non-linearly in the payoff. We run the full 50,000-path simulation at three scenarios:

Scenario	γ	μ	θ	Collar	Margin	PoD
Base (model)	0.163	9.2%	2.13%	0.05%	2.00%	1.20%
Realistic central	0.130	8.5%	2.70%	0.30%	2.20%	16.89%
Adverse plausible	0.100	8.0%	3.00%	0.40%	2.50%	43.15%



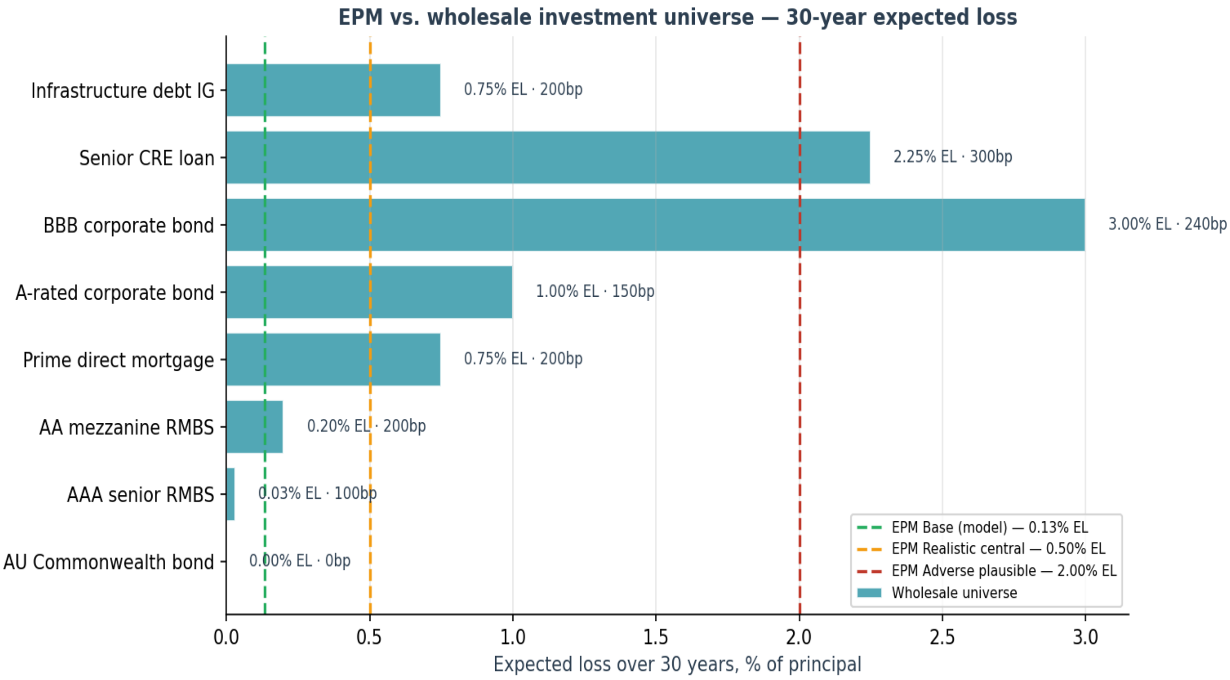
The "realistic central" scenario uses each parameter at roughly its empirical midpoint or forward-looking mid-case: γ = 0.13 (slightly below MLE to reflect small-sample bias), μ = 8.5% (between AU super high and CAPE-implied), θ = 2.7% (between US and AU long-run averages), collar = 0.30% (BS central), margin = 2.2%. PoD = 16.9%.

The "adverse plausible" scenario is not a tail: each input is well inside the observed empirical range. γ = 0.10, μ = 8.0%, θ = 3.0%, collar = 0.40%, margin = 2.5%. PoD = 43.2%.

This is our headline finding: parameter uncertainty alone — before any model-structure uncertainty — generates a 1–45% PoD range across scenarios that cannot be rejected by the available data.

10. Comparison to wholesale investment universe

A wholesale funder assessing EPM against the existing universe of fixed-income products faces the question: at 200bp over cash, does EPM offer a coherent risk/return trade-off?



Asset	EL 30yr	Spread	Rating
AU Commonwealth bond	0.00%	0bp	AAA
AAA senior RMBS	0.03%	100bp	AAA
AA mezzanine RMBS	0.20%	200bp	AA
Prime direct mortgage	0.75%	200bp	A-equiv
A-rated corporate bond	1.00%	150bp	A
BBB corporate bond	3.00%	240bp	BBB
Senior CRE loan	2.25%	300bp	unrated
Infrastructure debt IG	0.75%	200bp	A/BBB

At base-case parameters, EPM is roughly AA-mezzanine-RMBS equivalent on EL with a much higher spread — genuinely attractive. At realistic-central parameters, EPM is prime-mortgage-equivalent on EL at the same 200bp — fairly priced. At adverse-plausible, EPM sits alongside BBB corporates and senior CRE debt on EL but the 200bp spread is thin for that profile.

EPM scenario	EL 30yr	Spread	Comment
Base (model)	0.13%	200bp	Premium vs AA RMBS
Realistic central	0.50%	200bp	Prime mortgage equivalent
Adverse plausible	2.00%	200bp	BBB/CRE-like; spread too thin

11. Recommendations

Report PoD as a range, not a point. Headline reporting should quote 1–20% central range and 20–45% adverse range, with the base case preserved as "model as written" for traceability.

Anchor capital and reinsurance sizing on the adverse-plausible leg. Reinsurance attachment, wholesale-funder subordination, and internal solvency capital should be calibrated to the 43% PoD scenario, not the 1.2% base.

Adopt more conservative μ for pricing. Using μ = 8.0–8.5% for pricing (rather than 9.2%) gives a PoD in the 4–10% range — still competitive — without reliance on in-sample US equity performance continuing.

Replace the collar cost assumption. Move to 0.30% p.a. minimum (BS central). Re-test the product economics at 0.50% to stress-test for skew and roll slippage. If the collar cannot be bought at these levels, the product must price it in.

Explicitly disclose γ uncertainty. The mean-reversion assumption should be flagged in investor materials as the single most load-bearing modelling choice. Offer $\gamma = 0.10$ and $\gamma = 0$ sensitivity cases.

Track θ to AU cash, not US Fed Funds. Update the cash process to match the funding currency. A 50–150bp uplift to θ is empirically warranted.

Add a model-risk reserve. Beyond parameter risk, there remains residual model-structure risk (Gaussian shocks, no jumps, no regime switches, deterministic house prices). A 100–200bp reserve against scenario EL is appropriate.

12. Caveats & limitations

All analyses use 50,000-path Monte Carlo; Monte Carlo SE on PoD is ±0.1pp at base case, rising to ±0.3pp at the adverse-plausible scenario.

MLE on 36 annual observations is low-powered; confidence intervals are wide and the likelihood surface is flat in γ.

The analysis assumes the Shevchenko GBM+MeanRev specification is correct. Alternative specifications (Heston, jump-diffusion, regime-switching) are outside scope and represent additional model-structure risk.

Asset-class EL comparisons are based on long-run historical studies (S&P Global CreditPro, RBA FSR, Moody's Default and Loss Rates) and are themselves estimates rather than measurements.

House prices remain deterministic in v14c-003 by design — run-off mechanism eliminates market-value path dependence at maturity. This is an intentional product feature, not an omission.

— End of paper —